

National University of Computer and Emerging Sciences



Data Retrieval & Set Operations”



1. SELECT-FROM-WHERE

Select from where is equivalent to projection and selection in Relational Algebra, it will give output in form of a table. The most basic select statement includes Select and from clause, and it will retrieve all columns and rows from the table.

We will use the following schema and database for the examples. Find the queries for this database in InLab3Practice.sql and start practicing.

Students	StudentID	StudentName	StudentBatch	CGPA
	1	Ali	2013	3.3
	2	Aysha	2013	4
	3	Ahmed	2013	2.2

Instructors	InstructorID	InstructorsName
	1	Zafar
	2	Sadia
	3	Saima

Courses	CourseID	CourseName	CourseCreditHours	InstructorID
	1	Computer Programming	3	1
	2	Computer Organization	3	2
	3	Computer Programmi...	1	NULL

Registration	StudentID	CourseID	GPA
	1	1	3
	1	3	3
	2	2	0

Most Basic Select:

Retrieve data from table. Operator * after select means that all columns will be retrieved.

Syntax:

```
SELECT *
FROM <tableName>
```

Try this

```
select * from students
```

Results

	StudentID	StudentName	StudentBatch	CGPA
1	1	Ali	2013	3.3
2	2	Aysha	2013	4
3	3	Ahmed	2013	2.2



Retrieving Certain Columns from Select

To retrieve only certain columns give a comma separated list of those columns after Select keyword

Syntax:

SELECT ColumnX, ColumnY, ColumnZ

FROM <tableName>

Try this

```
Select Course Name, CourseCreditHours  
from courses
```

Results

	CourseName	CourseCreditHours
1	Computer Programming	3
2	Computer Organization	3
3	Computer Programming Lab	1

Retrieving Certain Rows from SELECT - WHERE CLAUSE

Rows can be filtered in SQL using WHERE clause. Rows that fulfill where clause conditions will be projected in result. Where clause can put condition on original columns of tables mentioned in from clause. Also, observe the use of Like operator in where clause.

Syntax:

SELECT *

FROM <tableName>

where <conditions>

Try this

```
Select CourseName, CourseCreditHours  
from courses  
where CourseName like '%Programming%' and CourseCreditHours >= 1
```

Results

	CourseName	CourseCreditHours
1	Computer Programming	3
2	Computer Programming Lab	1



Like Operator Scenarios

WHERE CourseName LIKE 'C%'	Finds any values that start with "C"
WHERE CourseName LIKE '%C'	Finds any values that end with "C"
WHERE CourseName LIKE '%Co%'	Finds any values that have "Co" in any position
WHERE CourseName LIKE '_r%'	Finds any values that have "r" in the second position
WHERE CourseName LIKE 'C_%'	Finds any values that start with "C" and are at least 2 characters in length
WHERE CourseName LIKE 'C__%'	Finds any values that start with "C" and are at least 3 characters in length
WHERE CourseName LIKE 'C%r'	Finds any values that start with "C" and ends with "r"

NOTE: % is referred to as **wildcard**.

Renaming Resulting Column

You can rename a column in result by using AS keyword also called Alias. The scope of this renaming is only to that select query, this is useful in joining where more than one table have same column names.

Syntax:

```
SELECT ColumnX as X , ColumnY as Y, ColumnZ
FROM <tableName> as Table1
```

Try this

```
select StudentName AS StudentFirstName
, CGPA AS [Cumulative CGPA]
from students AS StudentsTable
```

Results

	StudentFirstName	Cumulative CGPA
1	Ali	3.3
2	Aysha	4
3	Ahmed	2.2

SQL Server Built-in Functions

Sql Server has many built-in functions which can be used for different purposes.

For example:

- 1) GETDATE Returns the current database system date and time
- 2) CURRENT_TIMESTAMP Returns the current date and time
- 3) SUBSTRING Extracts some characters from a string

Syntax:

- 1) SELECT GETDATE();
- 3) SELECT CURRENT_TIMESTAMP;
- 2) SELECT SUBSTRING(columnName, startposition, substringlength) AS alias FROM <tableName>;



Try to explore as many string and data functions through this link:
https://www.w3schools.com/sql/sql_ref_sqlserver.asp

2. Order by Clause

Order by clause is used to arrange the rows in ascending or descending order of one or more columns

Syntax:

```
SELECT ColumnX as X, ColumnY as Y, ColumnZ  
FROM <tableName> as Table1  
ORDER BY ColumnX asc/desc, ColumnZ asc/desc
```

Try this

```
select StudentName AS StudentFirstName  
, CGPA AS [Cumulative CGPA]  
from students AS StudentsTable  
order by CGPA desc
```

Results

	StudentFirstName	Cumulative CGPA
1	Aysha	4
2	Ali	3.3
3	Ahmed	2.2

TOP Clause

Top n clause will give you first n rows from result instead of all the rows.

Syntax:

```
SELECT TOP <n> *  
FROM <tableName>  
WHERE <conditions>  
ORDER BY <column Name> asc/desc
```

Try this

```
SQLQuery7.sql - (local)\...Admin (55))* SQLQuery6.sql - (local)\...Adm  
select top 1 StudentName AS StudentFirstName  
, CGPA AS [Cumulative CGPA]  
from students AS StudentsTable  
order by CGPA desc
```

	StudentFirstName	Cumulative CGPA
1	Aysha	4



3. Arithmetic Operations

Sql arithmetic operators are:

- + Addition
- - Subtraction
- / Division
- * Multiplication
- % Modulus

All operations can be performed on either single column or multiple columns

Syntax:

1. Apply operation on single columns

```
SELECT ColumnX, ColumnY + 100
FROM <tableName>
```

2. Apply operation on multiple columns

```
SELECT ColumnX, ColumnY + ColumnZ
FROM <tableName>
```

Replace + with other operators and try them out yourself.

```
select CourseId, CourseName, CourseCreditHours, CourseCreditHours + 1
AS UpdatedCourseCreditHours from Courses

select CourseId, CourseName, CourseCreditHours, CourseId + CourseCreditHours
AS UpdatedCourseCreditHours from Courses

select CourseId, CourseName, CourseCreditHours, CourseCreditHours * 2
AS UpdatedCourseCreditHours from Courses

select CourseId, CourseName, CourseCreditHours, CourseId * CourseCreditHours
AS UpdatedCourseCreditHours from Courses

select StudentId, StudentName, StudentBatch, CGPA, CGPA / 2
AS UpdatedCGPA from Students

select StudentId, StudentName, StudentBatch, CGPA / StudentId
AS UpdatedCGPA from Students
```

	CourseId	CourseName	CourseCreditHours	UpdatedCourseCreditHours
1	1	Computer Programming	3	4
2	2	Computer Organization and Assembly	3	4
3	3	Computer Programming Lab	1	2

	CourseId	CourseName	CourseCreditHours	UpdatedCourseCreditHours
1	1	Computer Programming	3	4
2	2	Computer Organization and Assembly	3	5
3	3	Computer Programming Lab	1	4

	CourseId	CourseName	CourseCreditHours	UpdatedCourseCreditHours
1	1	Computer Programming	3	6
2	2	Computer Organization and Assembly	3	6
3	3	Computer Programming Lab	1	2

	CourseId	CourseName	CourseCreditHours	UpdatedCourseCreditHours
1	1	Computer Programming	3	3
2	2	Computer Organization and Assembly	3	6
3	3	Computer Programming Lab	1	3

	StudentId	StudentName	StudentBatch	CGPA	UpdatedCGPA
1	1	Ali	2013	3.3	1.65
2	2	Aysha	2013	4	2
3	3	Ahmed	2013	2.2	1.1

	StudentId	StudentName	StudentBatch	UpdatedCGPA
1	1	Ali	2013	3.3
2	2	Aysha	2013	2
3	3	Ahmed	2013	0.733333333...



4. Set Operations

Result of two (or more) select queries can be combined using set operations such as UNION, INTERSECT, EXCEPT.

Syntax:

```
SELECT ColumnX, ColumnY  
FROM <tableName>
```

Union/Intersect/Except

```
SELECT ColumnX, ColumnY  
FROM <tableName>
```

NOTE: The output of first select query should have same number and type of column as of second select query.

Try this –Set operations

```
--List IDs of all the students that have not registered in any course  
select StudentID From Students  
except  
select StudentID from Registration
```

Results Messages

StudentID
3
4
5

```
---list ID of all the instructors that are taking some course  
Select InstructorID from Instructors  
Intersect  
Select InstructorID from Courses
```

Results Messages

InstructorID
1
2

```
--List all the Names of instructors and Students  
Select StudentName From Students  
Union  
Select InstructorsName From Instructors
```

Results Messages

StudentName
Ahmed
Ali
Aysha
Bilal
Sadia
Saima
Zafar



Try this - error to look out for in set operations

```
--Fails because Datatype of Corresponding Columns is not same
Select StudentName From Students
Union
Select StudentID From Students
```

Results Messages

Msg 245, Level 16, State 1, Line 1
Conversion failed when converting the varchar value 'Ali' to data type int.

```
--Fails because Number of Columns in 1st select query are not same as number of columns in 2nd select query
Select StudentName From Students
Union
Select StudentID, StudentName From Students
```

Messages

Msg 205, Level 16, State 1, Line 2
All queries combined using a UNION, INTERSECT or EXCEPT operator must have an equal number of expressions in their target lists.